

Data Science and Visualization

Unit I FOUNDATIONS OF DATA SCIENCE



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Unit I – Foundations of Data Science

Defining Data Science - The Data Science Process - Data Types and Attributes - Python Essentials: NumPy & Pandas - Data Loading and Inspection - Data Visualization Goals - Introduction to Matplotlib & Seaborn - Basic Plotting and Customization.

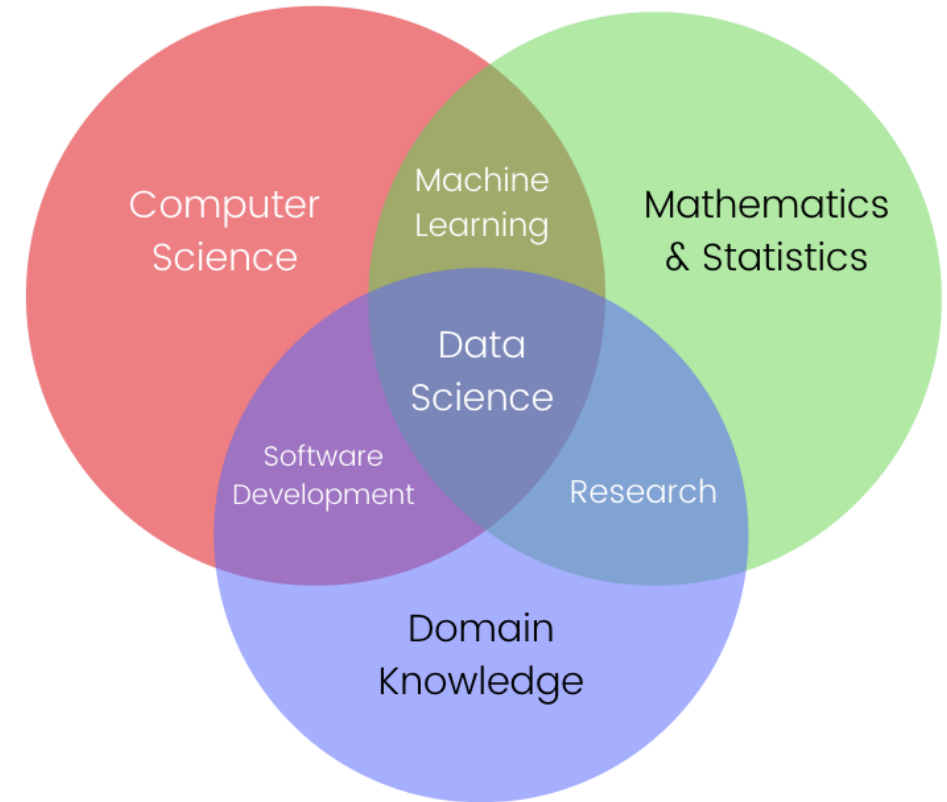
Unit I – Foundations of Data Science

Defining Data Science - The Data Science Process - Data Types and Attributes

Defining Data Science-Big Data context-Role of Data Science in AI/ML - The Data Science Process: Overview-key steps - Machine Learning definition and relationship with Data Science - Recognizing Different Types of Data - Structured, Unstructured - Introduction to Attributes/Features: Definition - Types -Categorical vs. Numerical

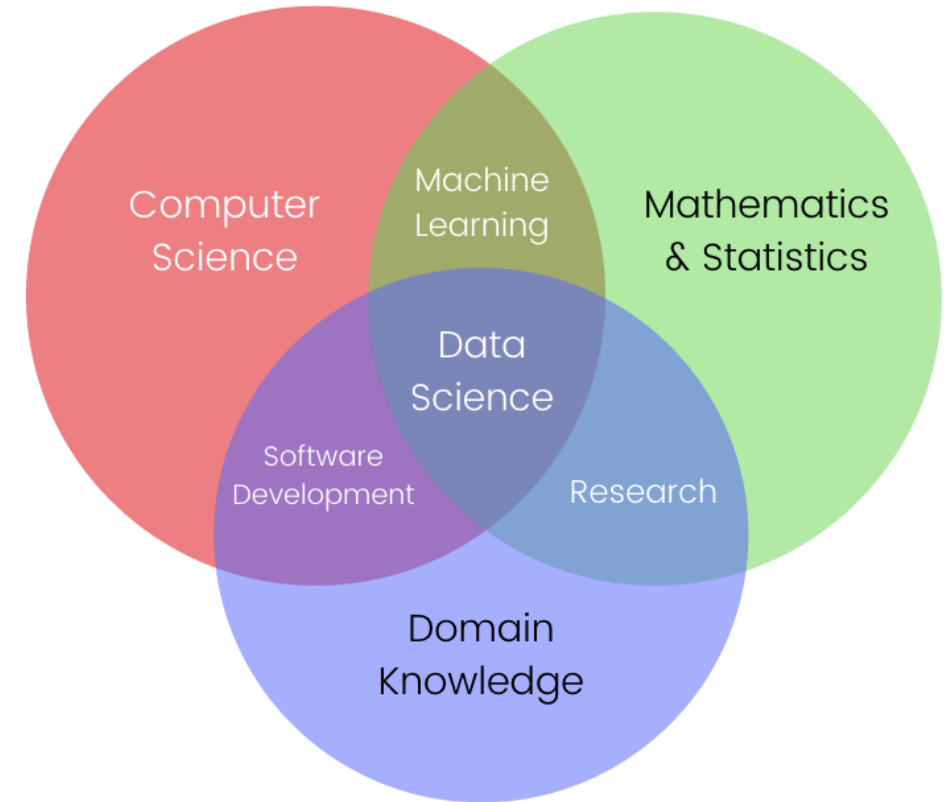
Defining Data Science

Data science is an interdisciplinary field that uses **scientific methods, algorithms, and programming** to **extract knowledge and insights** from large, complex datasets, combining statistics, machine learning, and domain expertise to **uncover patterns, build predictive models, and guide data-driven decisions** for various applications.



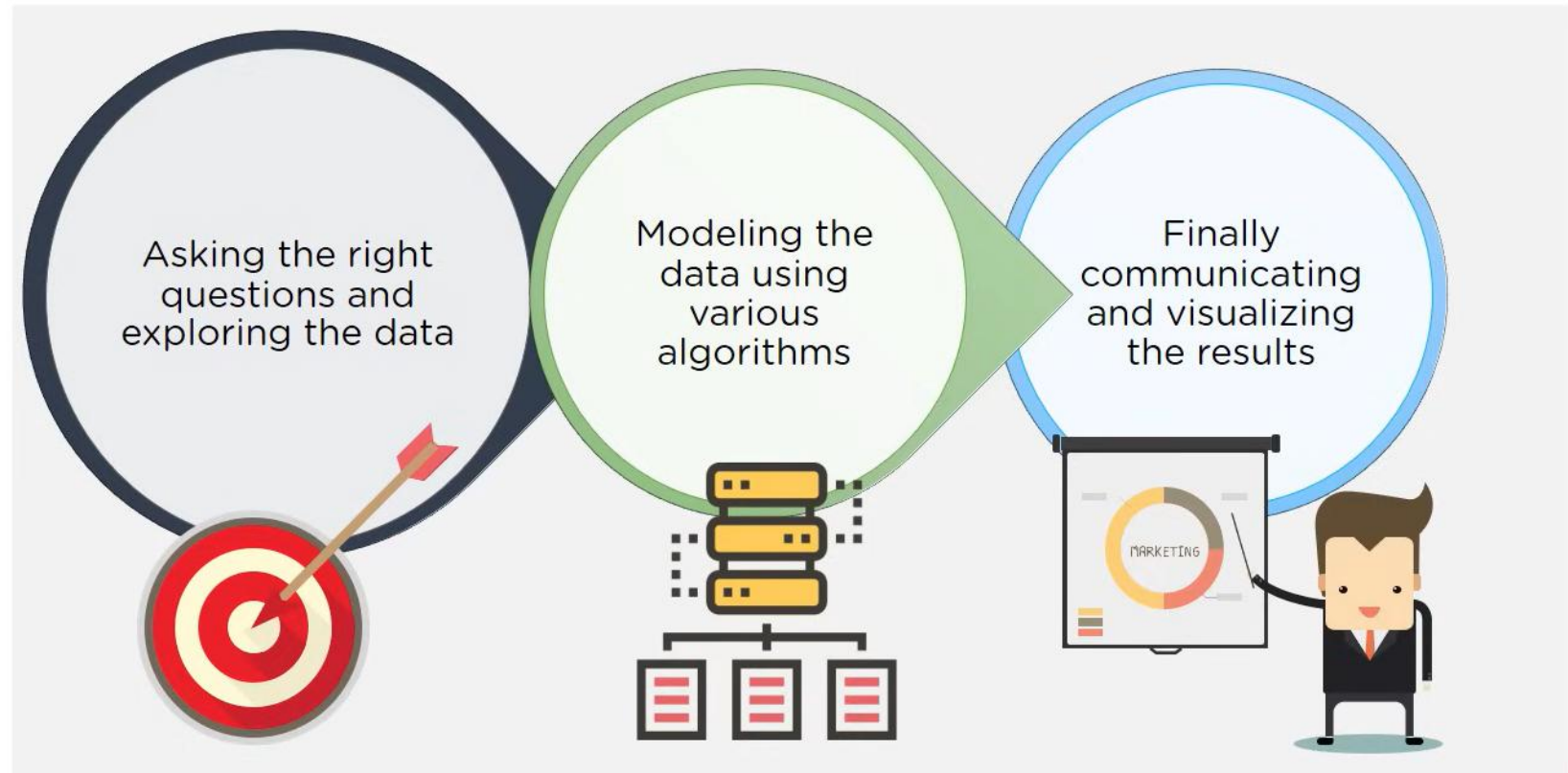
Defining Data Science

Data Science involves **collecting, cleaning, analyzing data, and presenting findings to solve problems, forecast trends, and automate processes** in fields from healthcare to finance.



What is Data Science?

So, Data Science or Data-driven Science is about:



Need For Data Science

So Data Science is mainly needed for:



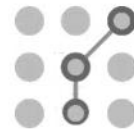
Better Decision Making

Whether A or B?



Predictive Analysis

What will happen next?



Pattern Discovery

Is there any hidden information in the data?

Data Science – Core Concepts

Multidisciplinary Approach: Blends math, statistics, computer science (AI/ML), and business/domain knowledge.

Knowledge Extraction: Aims to find actionable insights and hidden patterns from data.

Predictive Modeling: Uses algorithms to forecast future outcomes (e.g., customer behavior, sales).

Automation: Creates systems for real-time analysis, like fraud detection.

Data-Driven Decisions: Translates complex data into understandable information for better strategic choices.

Data Science – Key Applications

- **E-commerce:** Personalized recommendations, demand forecasting.
- **Healthcare:** Predicting disease, optimizing treatments.
- **Finance:** Fraud detection, risk assessment, market analysis.
- **Logistics:** Route optimization, delivery predictions.

Data science and artificial intelligence are not the same.

“Data science and artificial intelligence are two technologies that are transforming the world. While artificial intelligence powers data science operations, data science is not completely dependent on AI.

Data Science is leading the fourth industrial revolution.”

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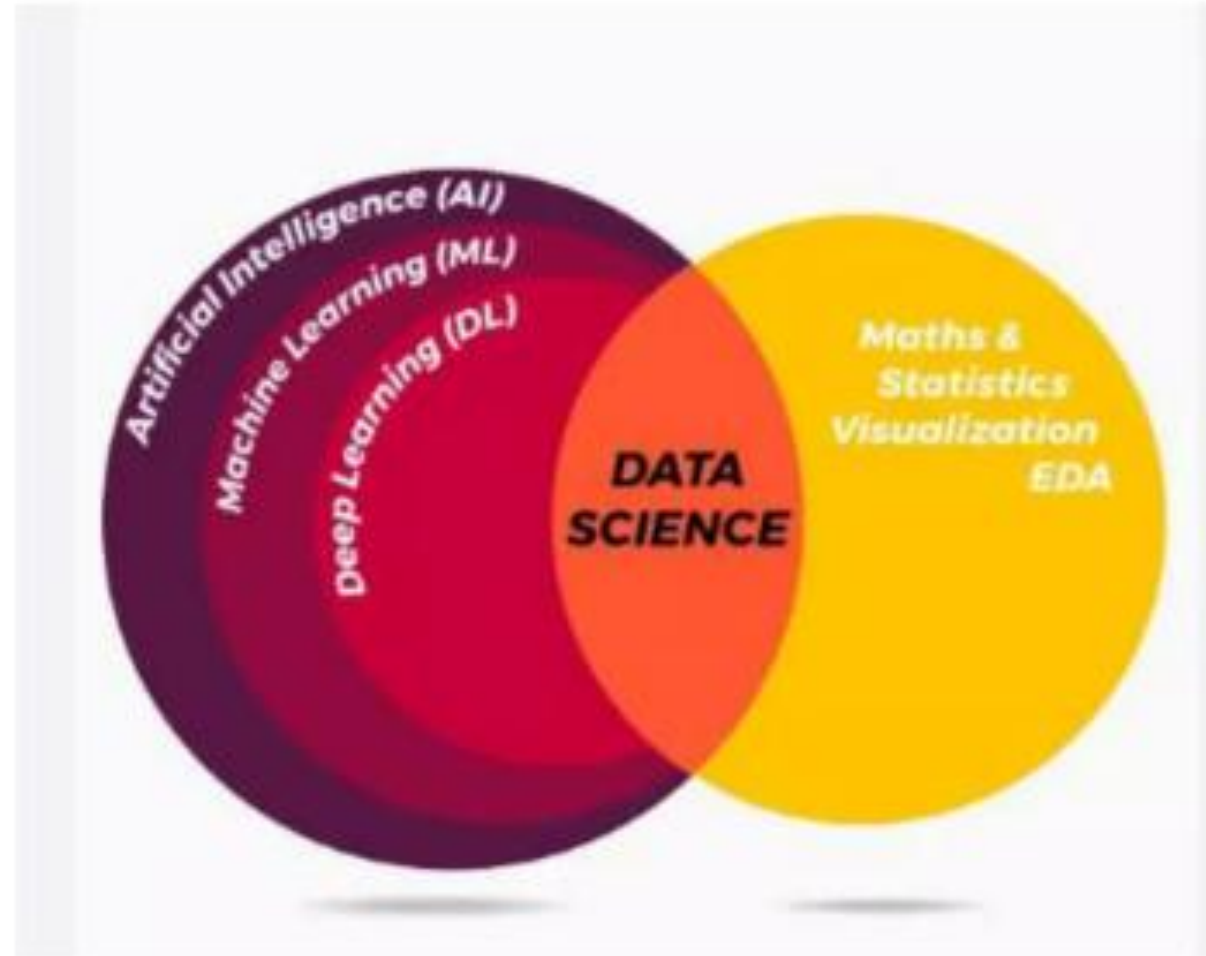


Data science also requires machine learning algorithms, which results in dependency on AI.

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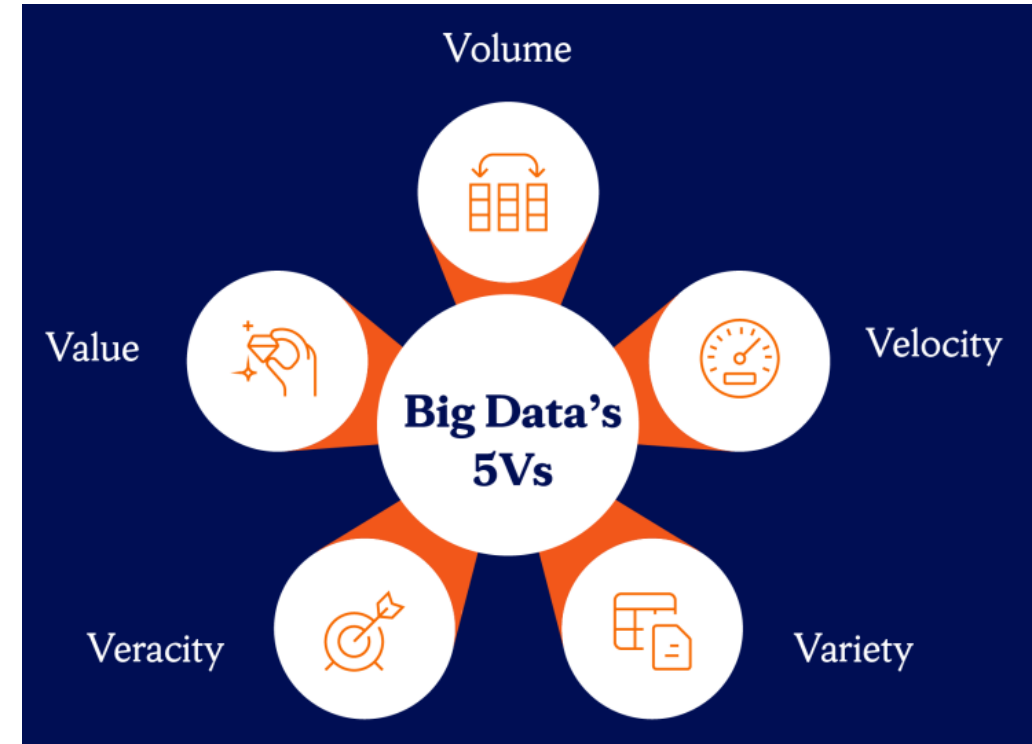


Comparison Between AI and Data Science



Data Science – Big Data Context

In the context of Big Data, Data Science is the discipline that uses statistics, programming, and machine learning to extract meaningful insights from massive, **complex datasets (Big Data)** that traditional tools can't handle, **characterized by the 3 Vs: Volume, Velocity, and Variety**, to enable data-driven decisions, predict trends, and optimize operations, turning raw information into actionable business value.



Big Data: The Challenge

Definition: Extremely large, fast-growing, and diverse datasets (structured, unstructured, semi-structured) from sources like social media, sensors, and transactions.

The 3 Vs:

- **Volume:** Huge amounts (terabytes to petabytes).
- **Velocity:** Data generated and needing analysis in near real-time.
- **Variety:** Different formats (text, images, logs, numbers).

The Problem: Traditional databases struggle to store, process, and analyze these huge, varied, and fast-moving data streams.

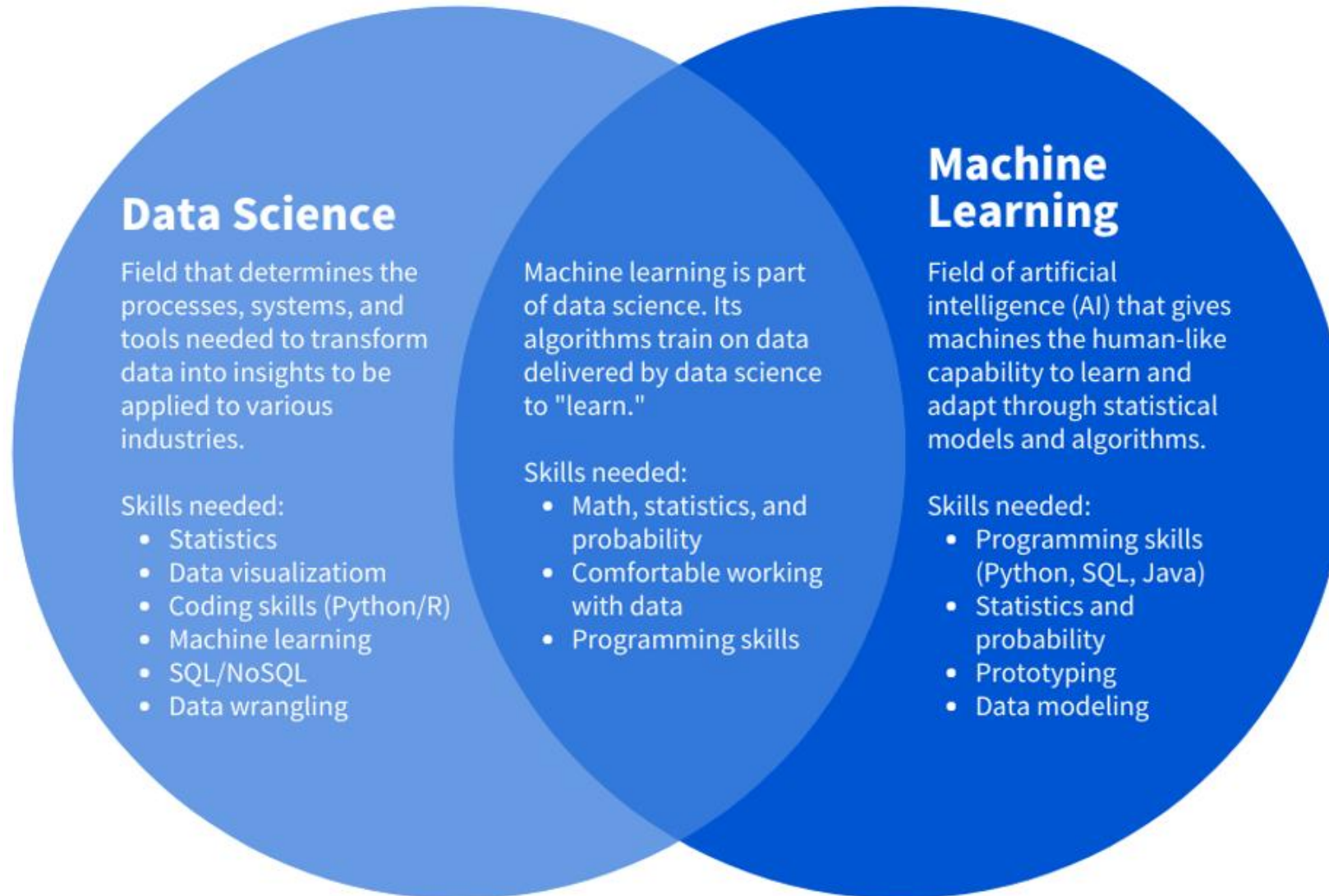
Data Science: The Solution

- **Role:** Applies scientific methods (math, stats, algorithms) to find patterns, predict outcomes, and gain insights from Big Data.
- **Key Techniques:** Data mining, predictive modeling, visualization, machine learning.
- **Goal:** Transform raw data into valuable, actionable intelligence for businesses.

Big Data – DS: The Interconnection

- Big Data provides the raw material; Data Science provides the tools and expertise to refine it.
- Data Scientists use specialized tools (Python, R, Big Data platforms like Spark/Hadoop) to handle Big Data's scale and complexity.
- **Examples:** Personalizing e-commerce recommendations, detecting fraud in financial transactions, optimizing logistics, understanding customer behavior.
- Big Data is the "**what**" (the massive dataset).
- Data Science is the "**how**" (the process of making sense of it).

Role of Data Science in AI/ML



Role of Data Science in AI/ML

- Data science is the foundation for AI/ML, providing the data, techniques, and evaluation to build intelligent systems.
- It involves collecting, cleaning, and analyzing data, **while ML uses those insights to create algorithms that learn patterns, enabling AI to make predictions and automate tasks**, creating a symbiotic relationship where data science prepares the fuel and ML builds the engine, with data science evaluating performance and ensuring continuous improvement.

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Key Roles of Data Science in AI/ML

- **Data Preparation & Feature Engineering:** Data scientists gather, clean, and transform raw data, creating the high-quality datasets that ML models need to train effectively.
- **Model Training & Algorithm Selection:** They select appropriate ML algorithms (like regression, classification) and use data to train AI models to recognize patterns and make predictions.
- **Model Evaluation & Validation:** Data science techniques are used to rigorously test AI model performance, identify weaknesses, and ensure accuracy and reliability.
- **Insight Generation:** Data scientists extract actionable insights from data, informing the business goals and strategies that AI systems aim to achieve.
- **Feedback Loops & Iteration:** They establish feedback mechanisms to continuously feed new data into AI systems, allowing them to learn and improve over time.
- **Automation & Deployment:** Data science pipelines automate the entire process, from data ingestion to model deployment, making AI solutions scalable and efficient.

DS – ML : The Interdependence

- **Data Science Fuels AI:** Data science provides the "what" (insights) and the "how" (methods) for AI to function; without data science, AI lacks the necessary data and context.
- **AI Enhances Data Science:** AI (especially ML) serves as a powerful tool within data science, automating complex analysis, uncovering hidden patterns, and building predictive models that go beyond traditional statistics.

In essence, **data science is the broader discipline** of extracting knowledge from data, while **ML is a core technique** used by data scientists, and **AI is the overarching goal of creating intelligent systems** that learn and act, all working together in a cyclical process.

How does Data Science Work?



Collect Data

Raw data is gathered from various sources that explain the business problem



Analyze Data

Using various statistical analysis and machine learning approaches, data modelling is performed to get the optimum solutions that best explain the business problem



Insights

Actionable insights that will serve as a solution for the business problems gathered through data science

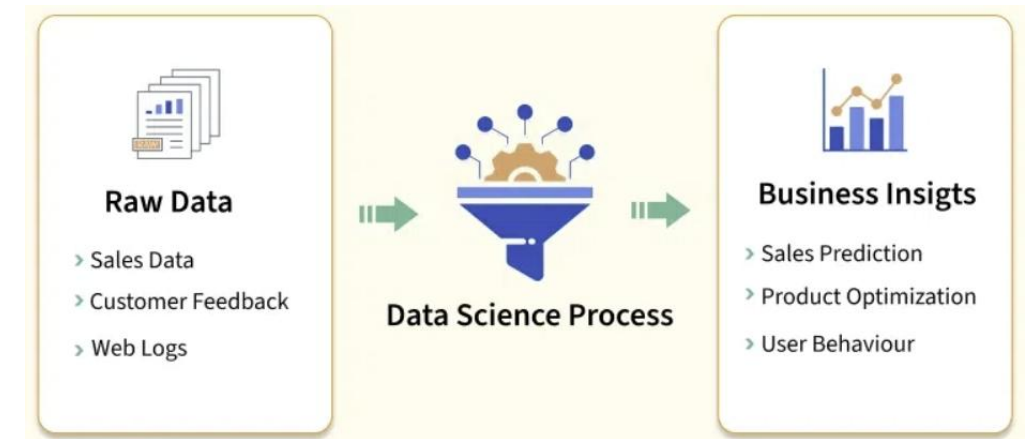
The Data Science Process



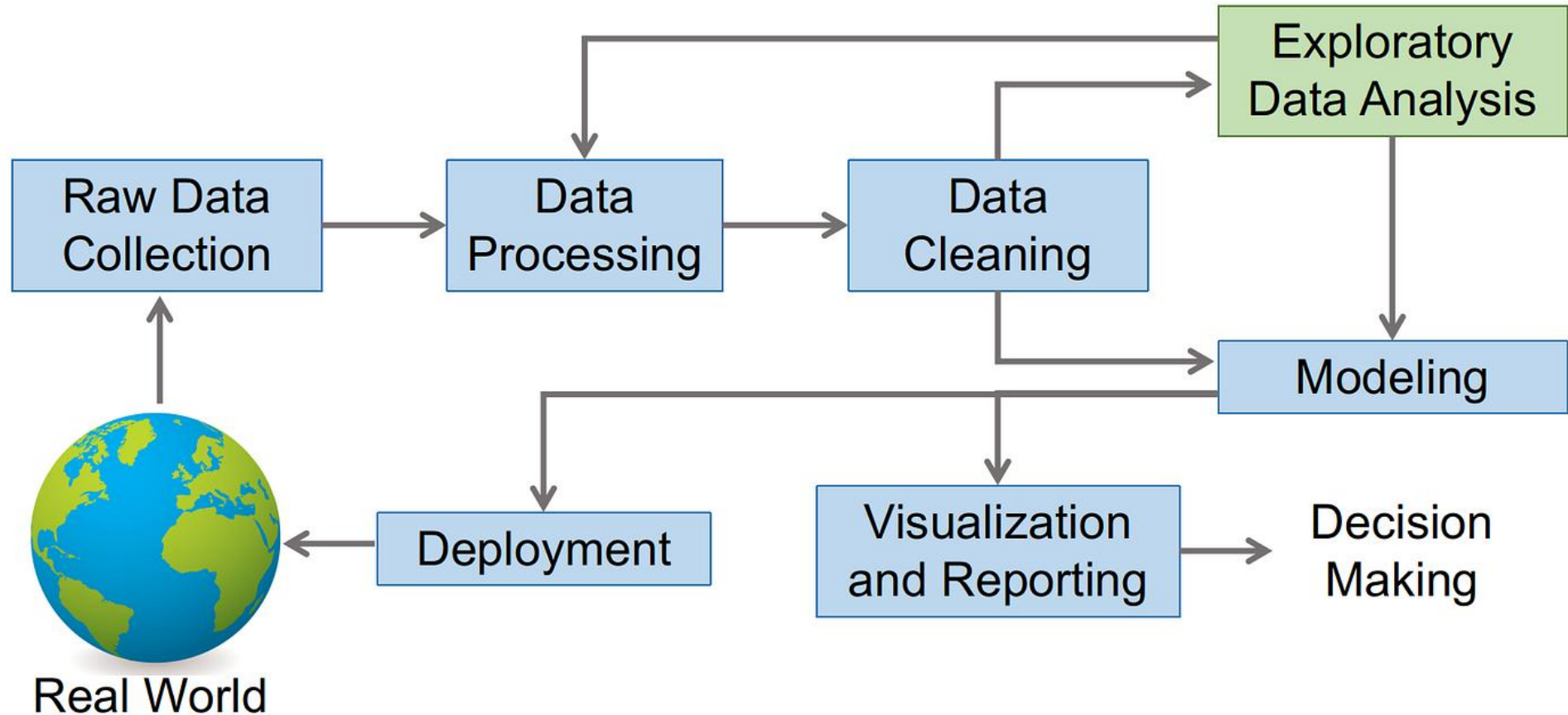
The Data Science Process - Overview

It processes raw data to address business challenges and predict future trends. For example, from large company datasets, data science can help answer questions like:

- What do customer want?
- How can we improve our services?
- What will the upcoming trend in sales?
- How much stock they need for upcoming festival?



The Data Science Process – Key Steps



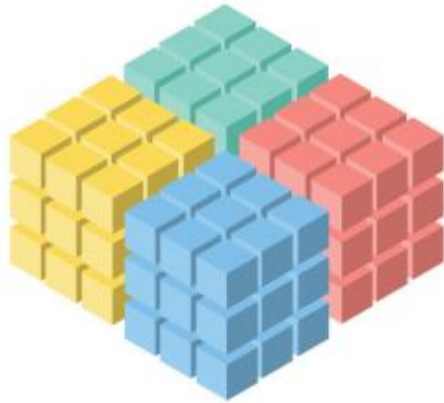
The Data Science Process

1. **Problem Definition:** Understand the business problem, its impact, the ultimate goals for addressing it, and the relevant project plan.
2. **Data Collection:** Gather data from various sources, such as databases, APIs, or web scraping.
3. **Data Processing:** Perform preliminary data processing, such as handling missing values, encoding categorical variables, and scaling numerical variables.
4. **Exploratory Data Analysis (EDA):** Explore the data using summary statistics and visualizations to better understand its characteristics, identify patterns, relationships, and outliers.
 - **Descriptive Statistics:** Calculate basic statistics, such as mean, median, mode, standard deviation, and variance, to summarize the data.
 - **Visualizations:** Create visualizations, such as bar charts, line charts, scatter plots, and histograms, to gain insights into the data.

The Data Science Process

5. **Data Cleaning:** Based on the insights from EDA, clean the data by addressing outliers, inconsistencies, and missing values.
6. **Modeling:** Use the cleaned and understood data to build and train machine learning models.
7. **Evaluation:** Assess the performance of the models using appropriate evaluation metrics, such as accuracy, precision, recall, and F1-score.
8. **Iteration:** If the model's performance is not satisfactory, return to the EDA step to refine the data understanding and cleaning process, and then rebuild and retrain the models.
9. **Reporting:** Communicate the results of the analysis, including the insights gained from EDA, the chosen models, and their performance.

Types of Data



Structured
Data



Semi-Structured
Data



Unstructured
Data

Structured Vs Unstructured data

Feature	Structured Data	Unstructured Data
Format	Schema-defined, tabular	No fixed format
Storage	Relational databases	Data lakes, NoSQL databases
Accessibility	Easily accessible via SQL	Requires advanced analytics
Examples	Transactional data, CRM systems	Text, images, videos
Processing	SQL, ETL pipelines	Machine Learning, NLP, Computer Vision
Uses	Financial reporting, inventory management	Sentiment analysis, image recognition, voice transcription

What is Data?

- A **data set** is a collection of objects and their attributes.
- An **attribute** is a property or characteristic of an object. **Examples:** eye color of a person, temperature, cost, etc. Attributes are also known as **variables, fields, characteristics, dimensions, or features.**
- A collection of attributes describe an **object**.
- Objects are also known as **records, points, cases, samples, entities, or instances.**

Objects

Attributes

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Attributes and Attribute Values

- Attribute values are **numbers or symbols** assigned to an **attribute** for a particular object.

Distinction between attributes and attribute values

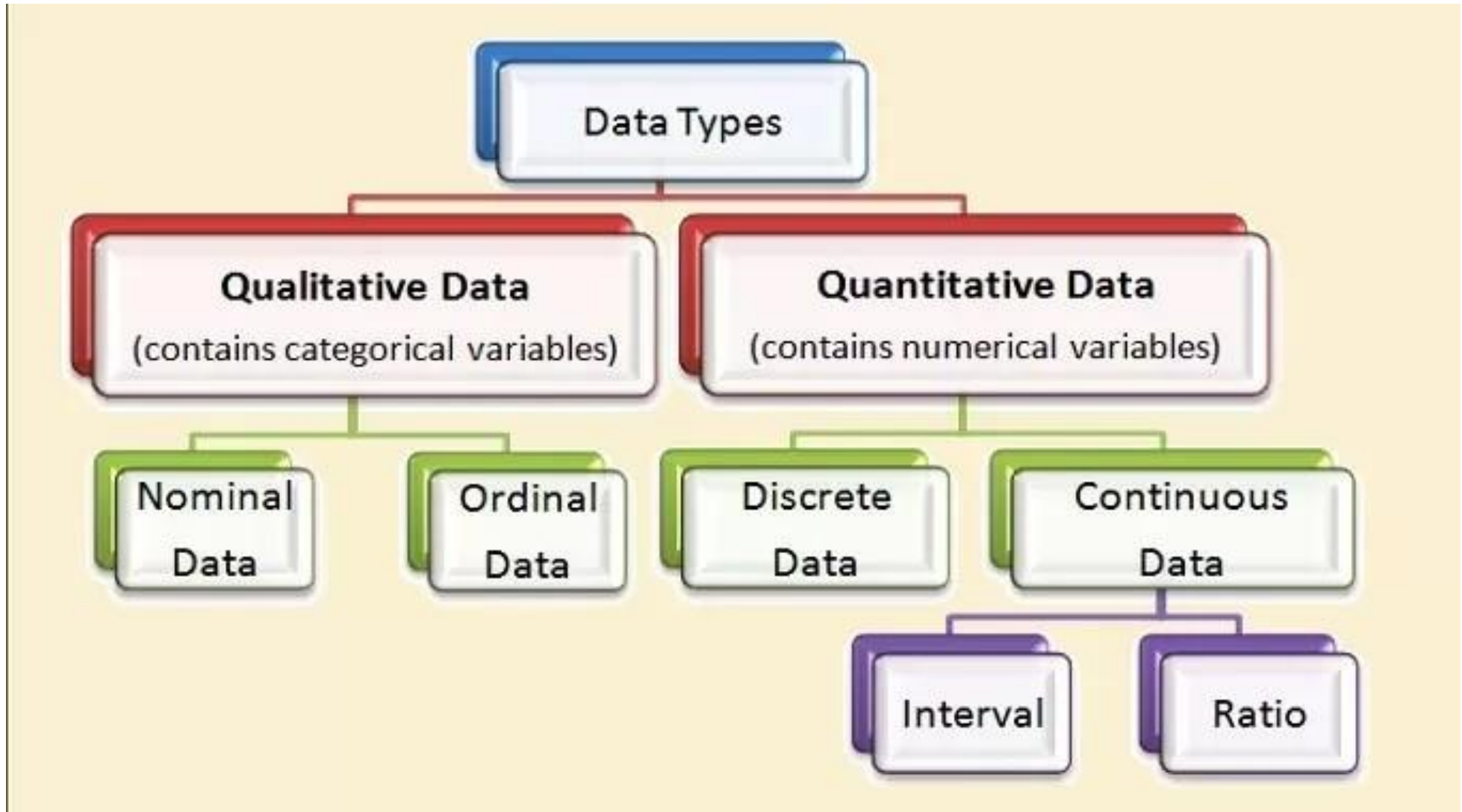
- Same attribute can be mapped to different attribute values.**
Example: height can be measured in feet or meters.
- Different attributes can be mapped to the same set of values.**
Example: Attribute values for ID and age are integers.
- But properties of attribute values can be different.

Attributes

Objects

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6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

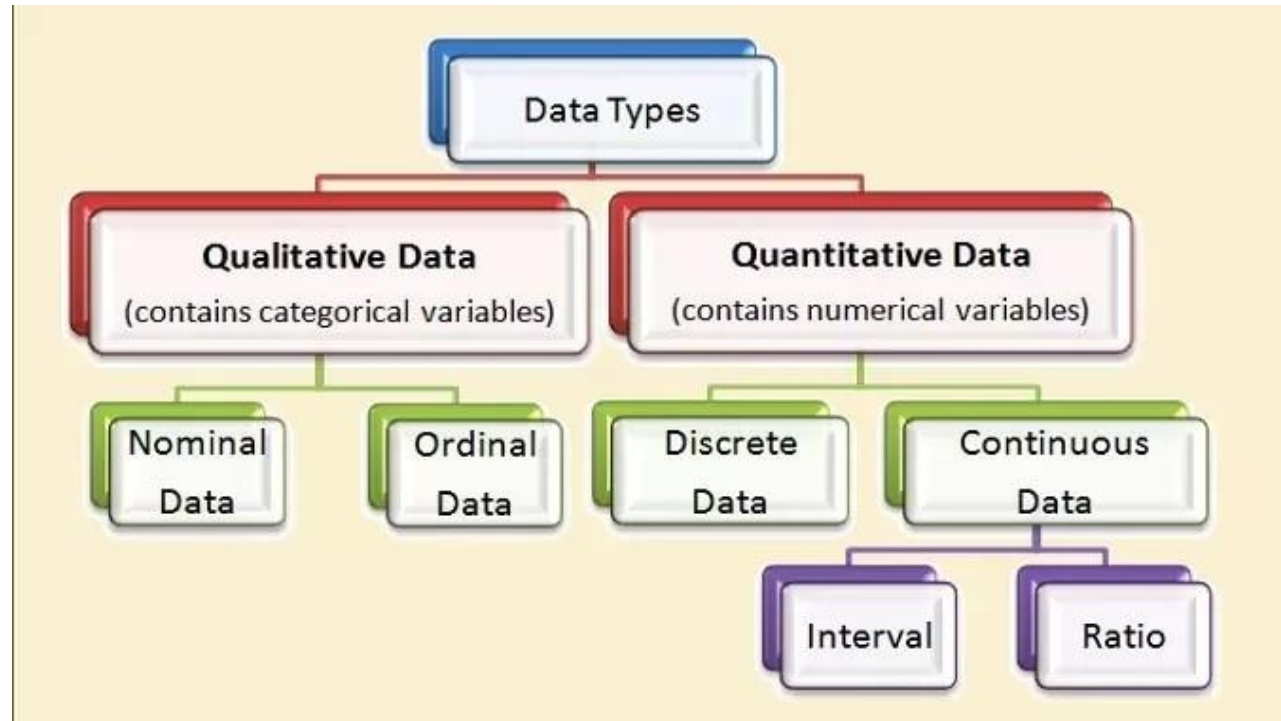
Data Attributes Categories in Data Analytics



Data Attributes Categories in Data Analytics

Data Types

- Data is classified based on the **nature of information** it represents.
- Data types help in selecting the correct method of analysis.
- Data is mainly divided into **qualitative data** and **quantitative data**.



Qualitative Data

- Qualitative data represents **categorical or descriptive information**.
- It is **non-numerical** and used to **describe characteristics**.
- It is classified into nominal data and ordinal data.

Nominal Data

- ✓ Nominal data consists of categories **without any specific order**.
- ✓ The values are used only for naming or labeling.
- ✓ **Example:** blood group, gender.

Ordinal Data

- ✓ Ordinal data consists of categories **with a meaningful order**.
- ✓ The difference between categories cannot be measured.
- ✓ **Example:** ranking, performance level (poor, good).

Quantitative Data

- Quantitative data represents **numerical and measurable values**.
- It answers questions such as **how much** or **how many**.
- It is classified into discrete data and continuous data.

Discrete Data

- ✓ Discrete data contains **countable values**.
- ✓ The values are usually whole numbers.
- ✓ **Example:** number of students, number of machines.

Continuous Data

- ✓ Continuous data can take **any value within a range**.
- ✓ It may include decimal values.
- ✓ **Example:** height, time.

Measurement Scales

- **Interval data** has equal intervals but **no true zero**.
- **Example:** temperature in Celsius.
- **Ratio data** has equal intervals and **a true zero**.
- **Example:** weight, distance.

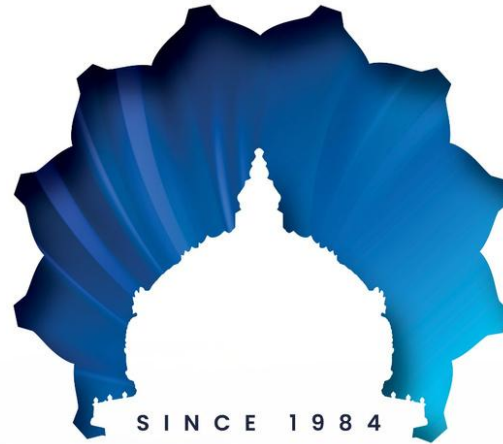
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